

smoothly, if at all, over time. Thus, before linkers were issued, many economists believed linkers would be the most boring of markets! The father of modern empirical asset pricing, Eugene Fama, argued in 1975 that the real rate was constant. He wrote the paper at a time when linkers were not available. Now that we have linkers, he was obviously wrong; Table 11.4 shows that real rates are volatile, and in fact approximately as volatile as nominal rates.

In Panel A of Figure 11.5, I plot real yields of five- and ten-year TIPS from the St. Louis Federal Reserve Bank from January 2003 to December 2011. On average, the real term structure is slightly upward sloping, but the real term spread between ten- and five-year maturities TIPS was approximately zero from 2006 to 2008 and was negative in 2008. The figure shows that real rates move around quite a bit: the five-year real rate moved from close to zero at the beginning of 2008 to over 4% that same year during some of the worst months of the financial crisis.

Panels B to D of Figure 11.5 graph the nominal Treasury yield and the TIPS yield for the five-, ten-, and twenty-year maturities, respectively. TIPS yields soared during the financial crisis as Treasury yields fell at the end of 2008. At this time there was a flight to quality as many investors flocked to safe Treasuries and sold other risky assets, including TIPS. Since both TIPS and Treasuries are backed by the full faith and credit of the U.S. government, both have the same near-zero default risk. Why would TIPS have sold off? Treasuries are the traditional tried-and-true, and Treasuries have roles in many places where TIPS do not (or where TIPS are rarely used) such as for collateral, meeting margin calls and, most of all, guaranteeing good liquidity. I discuss liquidity risk as a factor in TIPS below.

Panels A and B of Figure 11.5 show that the five-year real rate was negative and around -1% at December 2011. The ten-year TIPS yield (Panel C) was slightly below zero at this time. A negative real rate may seem surprising. Why would anyone want to buy a security and see it lose value in real terms over time? It is not a surprise from an economic point of view, however. While the older generation of models, which predate modern linkers markets, had trouble explaining the volatile real rates we see in the real world (newer models generate volatile real rates as investors’ assessments of risks change and firms’ financing and production opportunities change), they did accommodate the possibility of negative real rates.

In production models, the real rate represents how much future output is worth today: a real rate of 2% means that one widget today is worth 1.02 widgets next year. Usually this is positive because consumers are willing to forego output today for greater output in the future. The economy expands, on average, and there are positive production opportunities. With negative real rates, consumers want to consume today because output in the future is expected to be lower. Thus negative real rates can reflect shrinking future output opportunities—and a negative real rate is the ultimate bearish signal. This is the *production* interpretation of the real rate.